

MATH3303 – Lecture 20

Units, Group of Units, Finite Integral Domains, and Characteristic

Review Notes

Reference: Lee, Abstract Algebra, Sections 8.4–8.5

Overview

This lecture continues the study of rings, focusing on:

- **Units** – elements with multiplicative inverses
- **Uniqueness of multiplicative inverses**
- **Left/right multiplicative inverses** – and why they may differ in infinite rings
- **The group of units** $R^\times = U(R)$
- **Finite integral domains are fields** – a fundamental result
- **Characteristic of a ring** – the smallest n with $n \cdot r = 0$ for all r

Connection to Lecture 19. Last lecture introduced rings, subrings, zero divisors, integral domains, and fields. This lecture refines those ideas by studying which elements are invertible (units) and a fundamental numerical invariant (the characteristic).

1 Units in a Ring

1.1 Definition of a Multiplicative Inverse

Definition 1.1: Multiplicative Inverse (from lecture)

Let R be a ring. If $a \in R$, then an element $b \in R$ is called a **multiplicative inverse** of a if

$$ab = ba = 1.$$

Remark 1.1: Notation

Equivalent terminology used in Lee (Definition 8.10): we say that $a \in R$ is a **unit** if it has a multiplicative inverse. The inverse, when it exists, is denoted a^{-1} .

1.2 Uniqueness of the Multiplicative Inverse

Lemma 1.1: Uniqueness of the multiplicative inverse (from lecture)

A multiplicative inverse is **unique** if it exists.

Proof. Let b be a multiplicative inverse of a , and suppose b' is another such element. Then

$$ab = ba = 1 \quad \text{and} \quad ab' = b'a = 1.$$

We compute

$$b = b \cdot 1 = b(ab') = (ba)b' = 1 \cdot b' = b'.$$

Hence $b = b'$. □

Remark 1.2: Why this matters

Uniqueness justifies the notation a^{-1} – since there is only one inverse, the symbol is well-defined. This mirrors the uniqueness of inverses in a group, which we proved much earlier in the course.

1.3 Units and the Set $R^\times$

Definition 1.2: Unit (from lecture)

An element $a \in R$ is said to be a **unit** if it has a multiplicative inverse. The set of all units is denoted

$$R^\times \quad (\text{equivalently, } U(R) \text{ in Lee}).$$

Formally,

$$R^\times = \{r \in R : \exists s \in R \text{ such that } rs = sr = 1\}.$$

Example 1.1: Units of $\mathbb{Z}$ (Lee Example 8.25)

The only units of $\mathbb{Z}$ are ± 1 : indeed $1 \cdot 1 = 1$ and $(-1)(-1) = 1$, but no other integer $n \neq \pm 1$ admits an integer m with $nm = 1$.

$$\mathbb{Z}^\times = \{1, -1\}.$$

Example 1.2: Units of $\mathbb{Z}_n$ (Lee Example 8.26)

$U(\mathbb{Z}_n) = U(n) = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}$.

For instance:

- $U(\mathbb{Z}_6) = \{1, 5\}$
- $U(\mathbb{Z}_{10}) = \{1, 3, 7, 9\}$
- $U(\mathbb{Z}_{12}) = \{1, 5, 7, 11\}$

Example 1.3: Units of $M_n(\mathbb{R})$ (Lee Example 8.24)

By definition, $U(M_n(\mathbb{R})) = GL_n(\mathbb{R})$ – the general linear group of invertible $n \times n$ real matrices. A matrix A is a unit if and only if $\det(A) \neq 0$.

Example 1.4: Units of a field (Lee Example 8.27)

Every element other than 0 in $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ is a unit. In fact, this is the *defining property* of a field:

$$F \text{ is a field} \iff F \text{ is a commutative ring with } 1 \neq 0 \text{ and } F^\times = F \setminus \{0\}.$$

1.4 Left and Right Multiplicative Inverses

Definition 1.3: Left multiplicative inverse (from lecture)

An element $b \in R$ is called a **left multiplicative inverse** of $a \in R$ if

$$ba = 1.$$

Similarly, b is a **right multiplicative inverse** of a if

$$ab = 1.$$

Remark 1.3: Two-sided vs one-sided

A unit must have a *two-sided* inverse (both $ab = 1$ and $ba = 1$). In a noncommutative ring, an element may have only a left or only a right inverse without being a unit.

Finite rings – one-sided implies two-sided

Theorem 1.1: Exercise: Finite rings (from lecture)

In a **finite** ring with identity:

$$(1) \quad ba = 1 \implies ab = 1.$$

$$(2) \quad ab = 1 \implies ba = 1.$$

That is, in a finite ring, a one-sided inverse is automatically a two-sided inverse.

Proof sketch. Suppose $ba = 1$. Define $\varphi : R \rightarrow R$ by $\varphi(r) = ar$. If $ar_1 = ar_2$, then multiplying both sides on the left by b :

$$r_1 = b(ar_1) = b(ar_2) = r_2.$$

So φ is injective. Since R is finite, φ must also be surjective. Hence there exists $c \in R$ with $ac = 1$. Now

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c,$$

so $ab = ac = 1$ as well. □

Things that can go wrong in infinite rings

Remark 1.4: Infinite rings: left inverses need not be unique (from lecture)

If R is an **infinite** ring, a left multiplicative inverse is **NOT necessarily unique**.

Example 1.5: Infinite matrix ring (from lecture)

Consider $M_n(\mathbb{R})$ for infinite matrices – or more concretely, the ring of linear operators on an infinite-dimensional vector space (e.g., row-finite infinite matrices, or operators on $\mathbb{R}^\infty$).

Let S be the right-shift operator on sequences:

$$S(x_1, x_2, x_3, \dots) = (0, x_1, x_2, x_3, \dots).$$

Let T be the left-shift operator:

$$T(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots).$$

Then $TS = I$ (the identity), since shifting right and then left returns the original sequence. But $ST \neq I$ – because $ST(x_1, x_2, \dots) = (0, x_2, x_3, \dots)$, which loses the first coordinate.

So T is a left inverse of S , but S is *not* a unit. Moreover, any operator T' satisfying $T'S = I$ on the appropriate domain need not equal T , so left inverses are not unique.

2 The Group of Units

Lemma 2.1: $R^\times$ is a group (from lecture; Lee Theorem 8.8)

Let R be a ring with identity. Then $R^\times$ forms a group under multiplication, called the **group of units** of R .

Proof. We verify the group axioms.

(Identity) $1 \in R^\times$ since $1 \cdot 1 = 1$.

(Inverses) If $a \in R^\times$, then a^{-1} exists by definition. Moreover, a itself is the inverse of a^{-1} , so $a^{-1} \in R^\times$.

(Closure) Let $a, b \in R^\times$. We claim $ab \in R^\times$. Indeed,

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = a \cdot 1 \cdot a^{-1} = aa^{-1} = 1,$$

and similarly $(b^{-1}a^{-1})(ab) = 1$. So ab has inverse $b^{-1}a^{-1}$, hence $ab \in R^\times$.

(Associativity) Multiplication in R is associative, so $(ab)c = a(bc)$ for all $a, b, c \in R^\times$.

Thus $R^\times$ is a group. □

Remark 2.1: $R^\times$ may not be abelian

Even if R has identity, $R^\times$ need not be abelian. For example, $M_n(\mathbb{R})^\times = GL_n(\mathbb{R})$ is nonabelian for $n \geq 2$.

If R is a *commutative* ring, then $R^\times$ is an abelian group.

Example 2.1: Field of fractions (from lecture)

Let K be a field. Then

$$K[x]^\times = K^\times = K \setminus \{0\}.$$

That is, the units of the polynomial ring over a field are exactly the nonzero constants.

Why this is true. If $f(x), g(x) \in K[x]$ satisfy $f(x)g(x) = 1$, then taking degrees: $\deg(f) + \deg(g) = \deg(1) = 0$. Since degrees are nonnegative, $\deg(f) = \deg(g) = 0$, so both are nonzero constants. Conversely, every nonzero constant $c \in K^\times$ has inverse $c^{-1} \in K^\times \subseteq K[x]$. □

Example 2.2: More groups of units

- $\mathbb{Z}^\times = \{\pm 1\} \cong \mathbb{Z}_2$
- $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, an infinite abelian group
- $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$
- $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$
- $(\mathbb{Z} \oplus \mathbb{Z})^\times = \{\pm 1\} \times \{\pm 1\}$ (Klein four-group), by Lee Exercise 8.29
- $\mathbb{Z}_n^\times = U(n)$, which has order $\varphi(n)$ (Euler's totient)

3 Every Finite Integral Domain is a Field

This is a centerpiece of the lecture. It shows that finiteness collapses the gap between integral domains and fields.

Theorem 3.1: Every finite integral domain is a field (from lecture; Lee Theorem 8.10)

Every finite integral domain is a field.

Proof (from lecture). Suppose R is a finite integral domain. By definition, R is a commutative ring with $1 \neq 0$ and no zero divisors. We need to show that every nonzero element $r \in R$ is a unit.

Since R is finite, list its elements:

$$R = \{r_1, r_2, \dots, r_n\}.$$

Fix a nonzero $r \in R$. Multiply each element by r to form the list

$$rr_1, rr_2, \dots, rr_n.$$

Claim: all these products are distinct. Suppose $rr_i = rr_j$ for some $i \neq j$. Then

$$r(r_i - r_j) = 0.$$

Since $r \neq 0$ and R has no zero divisors, $r_i - r_j = 0$, i.e., $r_i = r_j$, contradicting $i \neq j$.

So $\{rr_1, rr_2, \dots, rr_n\}$ is a set of n distinct elements of R . But R has exactly n elements, so

$$\{rr_1, rr_2, \dots, rr_n\} = R.$$

In particular, $1 \in R$, so there exists some index i such that $rr_i = 1$. Since R is commutative, $r_i r = 1$ as well. Hence r is a unit with inverse $r_i = r^{-1}$.

So every nonzero $r \in R$ is a unit, proving R is a field. $\square$

Remark 3.1: Alternative proof (Lee's version, Theorem 8.10)

Lee's proof uses powers: for $r \neq 0$, the set $\{r^i : i > 0\}$ cannot have infinitely many distinct elements (since R is finite), so $r^i = r^j$ for some $i > j > 0$. Cancellation gives $r^{i-j} = 1$, so $r \cdot r^{i-j-1} = 1$, giving the inverse.

Both proofs work; the lecture's proof is slightly more elegant.

Example 3.1: Application: $\mathbb{Z}_n$ is a field $\iff n$ is prime (Lee Theorem 8.11)

If p is prime, $\mathbb{Z}_p$ is a finite commutative ring with $1 \neq 0$. It has no zero divisors (if $ab \equiv 0 \pmod{p}$, then $p \mid ab$, so $p \mid a$ or $p \mid b$). So $\mathbb{Z}_p$ is a finite integral domain, hence a field by the theorem above.

When p is prime, we often write $\mathbb{F}_p$ for $\mathbb{Z}_p$ to emphasise its field structure.

Example 3.2: $\mathbb{Z}$ is an integral domain but not a field

$\mathbb{Z}$ shows the theorem fails without finiteness: $\mathbb{Z}$ is an integral domain, but $2 \in \mathbb{Z}$ has no inverse, so $\mathbb{Z}$ is not a field. The theorem requires *finite* integral domains.

4 The Characteristic of a Ring

4.1 Definition

Definition 4.1: Characteristic (from lecture; Lee Definition 8.13)

Let R be a ring. The **characteristic** of R is the smallest positive integer n such that

$$n \cdot r = \underbrace{r + r + \cdots + r}_{n \text{ times}} = 0 \quad \text{for all } r \in R.$$

We denote this by $\text{char}(R)$.

If no such positive integer n exists, we say R has **characteristic zero**, written $\text{char}(R) = 0$.

Remark 4.1: Convention $\text{char} = 0$ versus $\text{char} = \infty$

Some texts say $\text{char} = \infty$; Lee and most modern references use $\text{char} = 0$. The choice is purely notational – both mean “no positive integer kills everything”.

4.2 Examples

Example 4.1: Characteristic zero rings (from lecture; Lee Example 8.32)

The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic zero. In $\mathbb{Z}$, for example, $n \cdot 1 = n \neq 0$ for any $n \geq 1$.

Example 4.2: Characteristic of $\mathbb{Z}_n$ (from lecture; Lee Example 8.31)

$\text{char}(\mathbb{Z}_n) = n$.

Indeed, for any $a \in \mathbb{Z}_n$, $n \cdot a = na \equiv 0 \pmod{n}$. And no smaller value works, since $m \cdot 1 = m \not\equiv 0 \pmod{n}$ for $1 \leq m < n$.

Example 4.3: Characteristic of a direct sum (from lecture)

$\text{char}(\mathbb{Z}_n \oplus \mathbb{Z}_m) = \text{lcm}(n, m)$.

Why: For $(a, b) \in \mathbb{Z}_n \oplus \mathbb{Z}_m$, we have $k \cdot (a, b) = (ka, kb) = (0, 0)$ in $\mathbb{Z}_n \oplus \mathbb{Z}_m$ iff $n \mid k$ and $m \mid k$, i.e., $\text{lcm}(n, m) \mid k$. The smallest such positive k is $\text{lcm}(n, m)$.

For instance:

- $\text{char}(\mathbb{Z}_2 \oplus \mathbb{Z}_3) = \text{lcm}(2, 3) = 6$
- $\text{char}(\mathbb{Z}_4 \oplus \mathbb{Z}_6) = \text{lcm}(4, 6) = 12$
- $\text{char}(\mathbb{Z}_2 \oplus \mathbb{Z}_2) = \text{lcm}(2, 2) = 2$

Example 4.4: Characteristic of $\mathbb{Z}_n[x]$ (from lecture)

$\text{char}(\mathbb{Z}_n[x]) = n$.

Why: For $f(x) = a_0 + a_1x + \cdots + a_kx^k \in \mathbb{Z}_n[x]$ with $a_i \in \mathbb{Z}_n$, we have $n \cdot a_i = 0$ in $\mathbb{Z}_n$, so $n \cdot f(x) = 0$. And no smaller value works (apply to the constant polynomial $f(x) = 1$).

Example 4.5: Characteristic of $\mathbb{Q}[x]$ (from lecture)

$\text{char}(\mathbb{Q}[x]) = 0$, since $\mathbb{Q}$ has characteristic 0.

Example 4.6: Characteristic of $M_n(\mathbb{Z}_p)$

$\text{char}(M_n(\mathbb{Z}_p)) = p$.

Why: For any matrix $A \in M_n(\mathbb{Z}_p)$, $p \cdot A$ multiplies each entry by p , which is 0 in $\mathbb{Z}_p$. And no smaller positive integer works (apply to the identity matrix).

4.3 Characteristic = Additive Order of 1**Lemma 4.1: Reduction to the identity (from lecture; Lee Theorem 8.13)**

Let R be a ring *with identity*. Then $\text{char}(R)$ equals the additive order of 1 in $(R, +)$:

$$\text{char}(R) = |1|_+$$

where $|1|_+$ is the smallest positive integer n such that $n \cdot 1 = 0$, or zero if no such n exists.

Proof (from lecture). **Case 1: 1 has infinite additive order.** Then there is no positive integer n with $n \cdot 1 = 0$, so $\text{char}(R) = 0$.

Case 2: 1 has additive order $n < \infty$. Then $n \cdot 1 = 0$. By definition of additive order, no smaller $1 \leq m < n$ satisfies $m \cdot 1 = 0$.

We show that $n \cdot a = 0$ for every $a \in R$:

$$n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}} = \underbrace{1 \cdot a + 1 \cdot a + \cdots + 1 \cdot a}_{n \text{ times}} = \underbrace{(1 + 1 + \cdots + 1)}_{n \text{ times}} \cdot a = (n \cdot 1) \cdot a = 0 \cdot a = 0.$$

So the additive order of *every* element a divides n , and certainly 1 achieves order n . Hence $\text{char}(R) = n$. $\square$

Remark 4.2: Practical implication

For rings with identity, you only need to look at the additive order of 1. This makes computing the characteristic very easy:

- $\text{char}(\mathbb{Z}) = 0$ because 1 has infinite additive order in $\mathbb{Z}$.
- $\text{char}(\mathbb{Z}_n) = n$ because 1 has additive order n in $\mathbb{Z}_n$.
- $\text{char}(\mathbb{Z}_n[x]) = n$ because the constant polynomial 1 has additive order n .

5 Summary and Hierarchy

5.1 Key Concepts at a Glance

Concept	Definition	Example
Unit	$\exists b$ with $ab = ba = 1$	± 1 in $\mathbb{Z}$
Left inverse	$\exists b$ with $ba = 1$	In infinite rings only
Right inverse	$\exists b$ with $ab = 1$	In infinite rings only
Group of units $R^\times$	Set of units, group under $\cdot$	$\text{GL}_n(\mathbb{R})$, $U(n)$
Characteristic 0	No positive n kills R	$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
Characteristic n	Smallest $n > 0$ with $n \cdot r = 0$	$\mathbb{Z}_n$, $M_n(\mathbb{Z}_p)$

5.2 Inclusion Chain

$$\boxed{\text{Field} \implies \text{Finite integral domain} \iff \text{Finite field}}$$

More generally:

$$\text{Field} \subset \text{Integral domain} \subset \text{Commutative ring with identity} \subset \text{Ring}$$

5.3 Quick Decision Diagram for Characteristic

Given a ring R with identity:

1. Compute the additive order n of 1 in $(R, +)$.
2. If $n = \infty$, then $\text{char}(R) = 0$.
3. Otherwise, $\text{char}(R) = n$.

6 Additional Key Theorems from the Textbook

Theorem 6.1: Units cannot be zero divisors (Lee Lemma 8.1)

Let R be a commutative ring with identity. Then a unit in R cannot be a zero divisor.

Proof. Suppose $a \in R^\times$ is also a zero divisor, so $ab = 0$ for some $b \neq 0$. Multiplying by a^{-1} on the left:

$$b = a^{-1}(ab) = a^{-1} \cdot 0 = 0,$$

contradicting $b \neq 0$. $\square$

Theorem 6.2: Every field is an integral domain (Lee Theorem 8.9)

Every field is an integral domain.

Proof. A field is commutative with $1 \neq 0$ by definition. If $a \neq 0$ in F , then a is a unit, hence not a zero divisor by the previous theorem. So F has no zero divisors and is an integral domain. $\square$

Theorem 6.3: Corollary: Unital subrings preserve characteristic (Lee Corollary 8.2)

Let R be a ring with identity. Then every unital subring of R has the same characteristic as R .

Proof. A unital subring contains the same identity element, which has the same additive order in the subring as in R . By Lemma 4.1, the characteristic is determined by this order. $\square$

Remark 6.1: Caution: non-unital subrings can differ

The corollary fails for non-unital subrings. For example, $R = \mathbb{Z}_6$ has $\text{char}(R) = 6$, but the subring $S = \{0, 2, 4\}$ has $\text{char}(S) = 3$ – because S does not contain the identity of $\mathbb{Z}_6$.

Theorem 6.4: Characteristic of an integral domain (Lee Theorem 8.15)

The characteristic of an integral domain is either 0 or a prime.

Proof. Let R be an integral domain with $\text{char}(R) = n > 0$. If $n = 1$, then $1 \cdot 1 = 0$, so $1 = 0$, contradicting the definition of an integral domain.

Suppose $n = kl$ with $1 < k, l < n$. Then

$$0 = n \cdot 1 = (kl) \cdot 1 = (k \cdot 1)(l \cdot 1).$$

Since R is an integral domain, either $k \cdot 1 = 0$ or $l \cdot 1 = 0$. But this contradicts the minimality of n as the additive order of 1. So n has no proper factorization and must be prime. $\square$

Theorem 6.5: Freshman's Dream (Lee Theorem 8.14)

Let R be a commutative ring of prime characteristic p . Then for any $a, b \in R$:

$$(a + b)^p = a^p + b^p.$$

Proof idea. By the binomial theorem,

$$(a + b)^p = \sum_{k=0}^p \binom{p}{k} a^{p-k} b^k.$$

For $1 \leq k \leq p-1$, the binomial coefficient $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ is divisible by p (since p is prime and does not divide $k!$ or $(p-k)!$). Since $\text{char}(R) = p$, these middle terms vanish, leaving only $a^p + b^p$. $\square$

7 Worked Examples

Worked Example 1: Find the group of units of $\mathbb{Z}_{12}$

We list $U(12) = \{a \in \mathbb{Z}_{12} : \gcd(a, 12) = 1\}$.

Checking $1, 2, \dots, 11$:

- $\gcd(1, 12) = 1$
- $\gcd(5, 12) = 1$
- $\gcd(7, 12) = 1$
- $\gcd(11, 12) = 1$
- All others share a factor with 12.

So $U(\mathbb{Z}_{12}) = \{1, 5, 7, 11\}$, which has order $\varphi(12) = 4$.

Inverses:

- $1^{-1} = 1$
- $5 \cdot 5 = 25 = 12 + 13 = 1$, so $5^{-1} = 5$
- $7 \cdot 7 = 49 = 4 \cdot 12 + 1 = 1$, so $7^{-1} = 7$
- $11 \cdot 11 = 121 = 10 \cdot 12 + 1 = 1$, so $11^{-1} = 11$

Every nonidentity element has order 2, so $U(\mathbb{Z}_{12}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (the Klein four-group).

Worked Example 2: Verify that $\mathbb{Z}_5$ is a field

$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ is a finite commutative ring with identity 1, and $1 \neq 0$.

No zero divisors: suppose $ab \equiv 0 \pmod{5}$ with $a, b \in \{1, 2, 3, 4\}$. Then $5 \mid ab$, and since 5 is prime, $5 \mid a$ or $5 \mid b$. But $a, b < 5$, so this is impossible.

So $\mathbb{Z}_5$ is a finite integral domain, hence a field by Theorem **Every finite integral domain is a field**. We can verify directly:

- $1^{-1} = 1$
- $2 \cdot 3 = 6 \equiv 1$, so $2^{-1} = 3$ and $3^{-1} = 2$
- $4 \cdot 4 = 16 \equiv 1$, so $4^{-1} = 4$

Worked Example 3: Compute the characteristic of $\mathbb{Z}_6 \oplus \mathbb{Z}_{10}$

For $(a, b) \in \mathbb{Z}_6 \oplus \mathbb{Z}_{10}$, we have $k \cdot (a, b) = (0, 0)$ iff $k \cdot a \equiv 0 \pmod{6}$ and $k \cdot b \equiv 0 \pmod{10}$ for all a, b .

Equivalently, $6 \mid k$ and $10 \mid k$, so $\text{lcm}(6, 10) \mid k$. The smallest such k is $\text{lcm}(6, 10) = 30$.

So $\text{char}(\mathbb{Z}_6 \oplus \mathbb{Z}_{10}) = 30$.

Alternatively, by Lemma **Reduction to the identity**, the additive order of $(1, 1)$ in $\mathbb{Z}_6 \oplus \mathbb{Z}_{10}$ is the answer. Order of 1 in $\mathbb{Z}_6$ is 6; order of 1 in $\mathbb{Z}_{10}$ is 10; so the order of $(1, 1)$ is $\text{lcm}(6, 10) = 30$.

Worked Example 4: Apply the Freshman's Dream in $\mathbb{Z}_3[x]$

Let $R = \mathbb{Z}_3$, so $\text{char}(R) = 3$, a prime. By Theorem **Freshman's Dream**,

$$(a + b)^3 = a^3 + b^3 \quad \text{for all } a, b \in \mathbb{Z}_3.$$

Verify with $a = 1, b = 1$ in $\mathbb{Z}_3$:

$$(1 + 1)^3 = 2^3 = 8 = 2 \pmod{3},$$

$$1^3 + 1^3 = 1 + 1 = 2 \pmod{3}.$$

Both sides equal 2.

8 Practice Problems

Units and the Group of Units

Problem 8.1: Units of $\mathbb{Z}_{18}$ (Lee Exercise 8.28(1))

For the ring $\mathbb{Z}_{18}$, determine which elements are units and which are zero divisors.

Problem 8.2: Units of $\mathbb{Z}_3 \oplus \mathbb{Z}_9$ (Lee Exercise 8.28(2))

For the ring $\mathbb{Z}_3 \oplus \mathbb{Z}_9$, determine which elements are units and which are zero divisors.

Problem 8.3: Units of direct sums (Lee Exercise 8.29)

Let R and S be rings with identity. Show that

$$U(R \oplus S) = U(R) \times U(S).$$

Problem 8.4: Units of $\mathbb{Z}_n$ (Lee Exercise 8.30)

Let $n \geq 2$ be a positive integer. Show that

$$U(\mathbb{Z}_n) = U(n) = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

Finite Integral Domains and Idempotents

Problem 8.5: Idempotents in an integral domain (Lee Exercise 8.31)

Show that every integral domain contains exactly two elements a satisfying $a^2 = a$.

Problem 8.6: When is a direct sum an integral domain? (Lee Exercise 8.32)

Let R and S be rings. Under precisely what circumstances is $R \oplus S$ an integral domain?

Problem 8.7: Cancellation in integral domains (Lee Exercise 8.35)

Let R be an integral domain. Suppose that $a, b \in R$ satisfy $a^{13} = b^{13}$ and $a^{10} = b^{10}$. Show that $a = b$.

Problem 8.8: Finite commutative ring without zero divisors (Lee Exercise 8.36)

Let R be a finite commutative ring having no zero divisors. Show that R is either $\{0\}$ or an integral domain.

Characteristic

Problem 8.9: Characteristic of $3\mathbb{Z}_{21}$ (Lee Exercise 8.37(1))

Find the characteristic of $3\mathbb{Z}_{21} = \{0, 3, 6, 9, 12, 15, 18\}$ (as a subring of $\mathbb{Z}_{21}$).

Problem 8.10: Characteristic of $\mathbb{R}[x]$ (Lee Exercise 8.37(2))

Find the characteristic of $\mathbb{R}[x]$.

Problem 8.11: Characteristic of $\mathbb{Z}_4 \oplus \mathbb{Z}_{10}$ (Lee Exercise 8.38(1))

Find the characteristic of $\mathbb{Z}_4 \oplus \mathbb{Z}_{10}$.

Problem 8.12: Characteristic of $M_2(\mathbb{Z}_3)$ (Lee Exercise 8.38(2))

Find the characteristic of the matrix ring $M_n(\mathbb{Z}_3)$ for $n = 2$.

Problem 8.13: Order of a finite integral domain (Lee Exercise 8.39)

Show that a finite integral domain R must have order p^n for some prime p and positive integer n .

Problem 8.14: Nilpotent + identity is a unit (Lee Exercise 8.41(1))

Let R be a commutative ring with identity, and suppose $a \in R$ satisfies $a^n = 0$ for some positive integer n . Show that $1 + a \in R^\times$.

Applications

Problem 8.15: Field of 4 elements (Lee Exercise 8.42)

Let $F = \{0, 1, a, b\}$ be a field with four elements. Write the addition and multiplication tables for F .

Solutions and Hints to Selected Problems

- **Units of $\mathbb{Z}_{18}$:**

$$U(\mathbb{Z}_{18}) = \{1, 5, 7, 11, 13, 17\}, \quad |U(\mathbb{Z}_{18})| = \varphi(18) = 6.$$

Zero divisors: all nonzero non-units, i.e., $\{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16\}$. For example, $2 \cdot 9 = 18 \equiv 0$, so 2 and 9 are zero divisors.

- **Units of $\mathbb{Z}_3 \oplus \mathbb{Z}_9$:** By the direct-sum result, $U(\mathbb{Z}_3 \oplus \mathbb{Z}_9) = U(\mathbb{Z}_3) \times U(\mathbb{Z}_9) = \{1, 2\} \times \{1, 2, 4, 5, 7, 8\}$, so $|U(\mathbb{Z}_3 \oplus \mathbb{Z}_9)| = 2 \cdot 6 = 12$.
- **Units of direct sums:** $(r, s) \in U(R \oplus S)$ iff there exists (r_1, s_1) with $(r, s)(r_1, s_1) = (rr_1, ss_1) = (1, 1)$ and similarly the other side. This requires $r \in U(R)$ and $s \in U(S)$.
- **Idempotents in an integral domain:** If $a^2 = a$, then $a(a - 1) = 0$. Since R has no zero divisors, $a = 0$ or $a = 1$.
- **When is a direct sum an integral domain?** $R \oplus S$ is an integral domain iff one of them is an integral domain and the other is $\{0\}$. Indeed, if both $R, S \neq \{0\}$, then $(1, 0)(0, 1) = (0, 0)$ gives zero divisors.
- **Cancellation in integral domains:** Compute $a^{13} \cdot a^{10} = b^{13} \cdot b^{10}$. So $a^{23} = b^{23}$. Combined with $a^{10} = b^{10}$ and $a^{13} = b^{13}$, use $\gcd(10, 13) = 1$: write $1 = 13 \cdot (-3) + 10 \cdot 4$. So $a \cdot (b^{13})^3 = a \cdot a^{39} = a^{40}$ and similarly $b \cdot a^{39} = b^{40}$. Conclude $a = b$ by cancellation.
- **Finite commutative ring without zero divisors:** Assume $R \neq \{0\}$, fix nonzero $a \in R$. The map $r \mapsto ar$ is injective (no zero divisors). On a finite set, injective $\implies$ surjective, so $ar = a$ for some r . Show r is a multiplicative identity, then check R is an integral domain.
- **Characteristic of $3\mathbb{Z}_{21}$:** Elements are $\{0, 3, 6, 9, 12, 15, 18\}$. The additive order of 3 in $\mathbb{Z}_{21}$ is 7 (since $7 \cdot 3 = 21 \equiv 0$). So $\text{char}(3\mathbb{Z}_{21}) = 7$.
- **Characteristic of $\mathbb{R}[x]$:** $\mathbb{R}$ has characteristic 0, so $\mathbb{R}[x]$ has characteristic 0.
- **Characteristic of $\mathbb{Z}_4 \oplus \mathbb{Z}_{10}$:** $\text{lcm}(4, 10) = 20$, so $\text{char} = 20$.
- **Characteristic of $M_2(\mathbb{Z}_3)$:** $\text{char} = 3$, the same as $\mathbb{Z}_3$.
- **Order of a finite integral domain:** A finite integral domain is a field by Theorem **Every finite integral domain is a field**. Its characteristic is prime p by Theorem **Characteristic of an integral domain**. Then $\mathbb{Z}_p \subseteq R$ (as additive subgroup), so R is a vector space over $\mathbb{Z}_p$ of some finite dimension n , giving $|R| = p^n$.

- **Nilpotent + identity is a unit:** If $a^n = 0$, then

$$(1 + a)(1 - a + a^2 - \cdots + (-1)^{n-1}a^{n-1}) = 1 - (-1)^na^n = 1.$$

So $1 + a$ is a unit with inverse $1 - a + a^2 - \cdots + (-1)^{n-1}a^{n-1}$.

- **Field of 4 elements ($\mathbb{F}_4$):**

Addition (characteristic 2, so $x + x = 0$):

+	0	1	a	b
0	0	1	a	b
1	1	0	b	a
a	a	b	0	1
b	b	a	1	0

Multiplication (must have $ab = 1$ since $F^\times$ is a group of order 3):

·	0	1	a	b
0	0	0	0	0
1	0	1	a	b
a	0	a	b	1
b	0	b	1	a

Here $b = a^2$ and $a^3 = 1$, so a generates the cyclic group $F^\times = \{1, a, a^2\} \cong \mathbb{Z}_3$.

Quick Review Checklist

- ☐ State the definition of a multiplicative inverse and explain why it is unique.
- ☐ Define a unit and the group of units $R^\times$.
- ☐ Distinguish between left, right, and two-sided multiplicative inverses.
- ☐ Prove that in a finite ring with identity, a one-sided inverse is automatically two-sided.
- ☐ Give an example of an infinite ring where left inverses are not unique.
- ☐ Prove that $R^\times$ is a group under multiplication.
- ☐ Compute $U(R)$ for $R = \mathbb{Z}, \mathbb{Z}_n, M_n(\mathbb{R}), K[x]$.
- ☐ State and prove that every finite integral domain is a field.
- ☐ Use this to conclude that $\mathbb{Z}_p$ is a field when p is prime.
- ☐ State the definition of the characteristic of a ring.
- ☐ Prove that for rings with identity, the characteristic equals the additive order of 1.
- ☐ Compute the characteristic of $\mathbb{Z}, \mathbb{Q}, \mathbb{Z}_n, \mathbb{Z}_n \oplus \mathbb{Z}_m, \mathbb{Z}_n[x], M_n(\mathbb{Z}_p)$.
- ☐ Prove that the characteristic of an integral domain is 0 or prime.
- ☐ State the Freshman's Dream and apply it in characteristic p .